

HAOS-IIP Biology Line A–D: Recoverability-Loss Detection in Toy Biological Interaction Systems

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Abstract

This numbered paper records the first HAOS-IIP Biology Line ladder: four toy biological interaction demonstrations built as experimental application layers on top of frozen HAOS-IIP. The sequence moves from an abstract synthetic gene regulatory network, to a microtubule-inspired cylindrical lattice, to a 3D multicellular tissue-like lattice, and finally to a 4D morphogenesis-style model defined as a 3D tissue state evolving through developmental time. In all four lines, lightweight HAOS-style proxy diagnostics detect recoverability loss before the configured visible-failure criterion becomes true. The results are: Line A $k_{\text{star}} = 0.7000$ before visible regulatory failure at 0.9333; Line B $k_{\text{star}} = 0.8333$ before visible lattice failure at 0.8667; Line C $k_{\text{star}} = 0.8000$ before visible tissue failure at 1.0000; and Line D $k_{\text{star}} = 0.8667$ before visible developmental failure at 0.9667. Each line is deterministic and includes a small robustness variation that preserves early detection. The claim is intentionally narrow: HAOS-IIP now has four toy biology-line proof-of-instrument artifacts showing early recoverability-loss detection across increasingly structured biological interaction systems. This is not real biological validation, not a tissue or morphogenesis theory, not a consciousness model, not quantum biology, and not a claim that HAOS-IIP explains biology.

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1 Status and Scope

This paper is a numbered release note for the HAOS-IIP Biology Line A–D ladder. It summarizes repository artifacts, not a new biological theory. The four demos are:

1. Biology Line A: synthetic gene regulatory network.
2. Biology Line B: microtubule-inspired cylindrical lattice.
3. Biology Line C: 3D multicellular tissue-like lattice.
4. Biology Line D: 4D morphogenesis-style development, meaning 3D tissue state through developmental time.

The common instrument question is:

At what perturbation level does a biological interaction structure stop being recoverable under perturbation?

The common result is that a lightweight HAOS-style diagnostic threshold, `k_star`, appears before the configured visible-failure threshold in all four demos.

1.1 Strict Boundaries

The following are excluded:

- no claim that HAOS-IIP explains biology;
- no claim of real biological validation;
- no molecular simulation claim;
- no developmental biology proof;
- no consciousness model;
- no quantum biology claim;
- no modification of frozen HAOS-IIP core, theory, telemetry, phase artifacts, or canonical notes.

The demos are application-layer experiments under `experiments/biology/`. Their metrics are lightweight external proxies aligned with HAOS-IIP recoverability logic. They are not frozen HAOS core metrics.

2 Release Map

Line	Artifact	Tag	Commit	
A	<code>experiments/biology/genetics/evolution-v0.1</code>	<code>bioevolution-v0.1</code>	09e0065, 626996d	lock
B	<code>experiments/biology/microbiology/hibnet-v0.1</code>	<code>microbiology-hibnet-v0.1</code>	77edf09	
C	<code>experiments/biology/tissue/c-v0.1</code>	<code>tissue-c-v0.1</code>	325e1d3	
D	<code>experiments/biology/morphology/se4d-v0.1</code>	<code>morphology-se4d-v0.1</code>	c3c0602	

3 Shared Diagnostic Form

Each line sweeps a perturbation parameter $p \in [0, 1]$, where $p = 0$ is the baseline and $p = 1$ is maximum perturbation. For each sampled perturbation level, a recoverability proxy is computed. The exact components differ by model, but every line follows the same diagnostic form.

3.1 Recoverability

Recoverability is a scalar in $[0, 1]$. It is a weighted proxy for whether the perturbed system remains close to the baseline or expected target organization. Depending on the line, it combines final-state match, component support, weighted-degree retention, signal propagation retention, spatial continuity, or trajectory coherence.

3.2 Delta Persistence

$$\Delta P_k = R_k - R_{k-1}.$$

Here R_k is recoverability at perturbation index k .

3.3 `k_star`

`k_star` is the first perturbation index where recoverability remains below a collapse threshold for the current sample plus a small sustained window. Lines B–D use:

```
collapse_threshold = 0.70,    sustain_steps = 2.
```

Line A uses `collapse_threshold = 0.65` with the same sustain setting.

3.4 Safety Margin and Confidence

Safety margin is the distance from the current perturbation level to p at `k_star`. At and beyond `k_star`, safety margin is zero or negative. Confidence is the magnitude of the strongest negative ΔP seen so far.

3.5 Visible Failure

Visible failure is intentionally stricter than recoverability collapse. It is a boolean criterion meant to represent obvious loss of functional or morphological organization. The central pass condition is:

$$p(\text{k_star}) < p(\text{first visible failure}).$$

4 Line A: Gene Network Demo

4.1 Artifact

- Path: `experiments/biology/gene_network_demo/`
- Runner: `run_gene_network_demo.py`
- Validation: `gene_network_validation.md`
- Results: `outputs/results.csv`

4.2 System

Line A uses a deterministic toy gene regulatory network with 12 genes, `G0` through `G11`. Each gene state is continuous in $[0, 1]$. Directed weighted edges represent regulatory influence: positive weights activate and negative weights inhibit. The network includes:

- one hub regulator, `G0`;
- a negative feedback loop involving `G1`, `G2`, and `G0`;
- a positive feedback motif between `G3` and `G4`;
- a fragile downstream branch `G8--G11`.

The deterministic dynamics are:

$$\begin{aligned}\text{input}_i(t) &= b_i + \sum_j W_{j,i} x_j(t), \\ x_i(t+1) &= (1-d)x_i(t) + d\sigma(\text{input}_i(t)),\end{aligned}$$

with clipping to $[0, 1]$.

4.3 Perturbation

The primary perturbation gradually weakens support edges feeding the fragile branch. Hub knock-down is implemented but is not the primary validation path.

4.4 Metric

Recoverability is:

$$R = 1 - \text{normalized_distance}(\text{perturbed final state}, \text{baseline final state}),$$

clipped to $[0, 1]$. The final state is the mean over the final trajectory window.

4.5 Result

Quantity	Value
Pass status	PASS
Baseline stable	True
<code>k_star</code> level	0.7000
First visible failure level	0.9333
Early detection	True
Fragile branch <code>k_star</code> level	0.4000
Minimum whole-network recoverability	0.602222

The robustness variation scales fragile-branch support edges by 1.03. It preserves early detection with `k_star` = 0.7000 and first visible failure at 0.8667.

5 Line B: Microtubule-Inspired Lattice Demo

5.1 Artifact

- Path: `experiments/biology/microtubule_lattice_demo/`
- Runner: `run_microtubule_lattice_demo.py`
- Validation: `microtubule_lattice_validation.md`
- Results: `outputs/results.csv`

5.2 System

Line B uses a coarse-grained deterministic cylindrical lattice inspired by a common microtubule geometry: 13 protofilaments and 24 dimers per protofilament, for 312 nodes. Each node represents one tubulin dimer position. This is not a molecularly realistic simulation.

Edge classes are:

- **longitudinal**: strongest support along a protofilament;
- **lateral**: medium circumferential support between protofilaments;
- **seam_or_diagonal**: weak deterministic offset support;
- **weak_support**: very weak longer-range support.

5.3 Perturbation

The primary perturbation is lateral weakening. Lateral edge strengths are multiplied by $1 - p$. A deterministic defect patch mode is implemented but is not the primary sweep.

5.4 Metric

Recoverability is:

$$R = 0.05C + 0.90D + 0.05S,$$

where C is largest connected component fraction, D is weighted-degree retention, and S is propagation retention.

5.5 Result

Quantity	Value
Pass status	PASS
Baseline stable	True
<code>k_star</code> level	0.8333
First visible failure level	0.8667
Early detection	True
Robustness variation	<code>longitudinal_strength_x_1.02</code>
Robustness early detection	True
Minimum recoverability	0.578094

6 Line C: 3D Tissue Demo

6.1 Artifact

- Path: `experiments/biology/tissue_3d_demo/`
- Runner: `run_tissue_3d_demo.py`
- Validation: `tissue_3d_validation.md`
- Results: `outputs/results.csv`

6.2 System

Line C uses a deterministic 3D multicellular tissue-like lattice with grid size $8 \times 8 \times 8$, for 512 cells. Each node is a cell with a scalar state in $[0, 1]$. Edge classes are:

- `local_neighbor`: 6-connected nearest-neighbor support;
- `diagonal_neighbor`: weak local support;
- `long_range_signal`: sparse deterministic coordination links.

6.3 Perturbation

The primary perturbation is a local lesion centered at $(3.5, 3.5, 3.5)$ with radius 4.1. Perturbation level p weakens edge support inside that spherical region. A secondary long-range signaling weakening mode is implemented but is not the primary sweep.

6.4 Metric

Recoverability is:

$$R = 0.15C + 0.45D + 0.25Q + 0.15S,$$

where C is largest component fraction, D is weighted-degree retention, Q is spatial-coherence retention, and S is propagation retention.

6.5 Result

Quantity	Value
Pass status	PASS
Baseline stable	True
<code>k_star</code> level	0.8000
First visible failure level	1.0000
Early detection	True
Robustness variation	<code>local_neighbor_strength_x_1.02</code>
Robustness early detection	True
Minimum recoverability	0.563310

7 Line D: 4D Morphogenesis Demo

7.1 Artifact

- Path: `experiments/biology/morphogenesis_4d_demo/`
- Runner: `run_morphogenesis_4d_demo.py`
- Validation: `morphogenesis_4d_validation.md`
- Results: `outputs/results.csv`

7.2 System

Line D adds developmental time to the 3D tissue idea. It uses an $8 \times 8 \times 8$ cell lattice with 24 developmental time steps. The target morphology is deterministic:

- early development: compact central form;
- mid development: axis polarity gradient;
- late development: core-shell/anterior-posterior pattern.

Edge classes are:

- `local_neighbor`;
- `diagonal_neighbor`;
- `developmental_signal`.

7.3 Perturbation

The primary perturbation is developmental drift. It delays and weakens target pull in a deterministic region during mid-to-late development and weakens developmental-signal links in that affected region. A local developmental lesion mode is implemented but is not the primary validation sweep.

7.4 Metric

Recoverability is:

$$R = 0.35F + 0.32T + 0.25I + 0.08Q,$$

where F is final morphology match, T is trajectory coherence, I is interaction-support retention, and Q is spatial-continuity retention.

7.5 Result

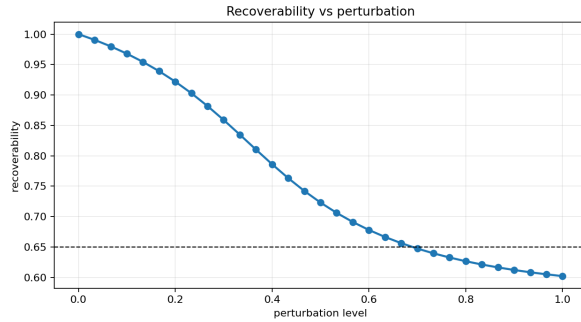
Quantity	Value
Pass status	PASS
Baseline stable	True
<code>k_star</code> level	0.8667
First visible failure level	0.9667
Early detection	True
Robustness variation	<code>target_weight_x_1_02</code>
Robustness early detection	True
Minimum recoverability	0.654913

8 Cross-Line Comparison

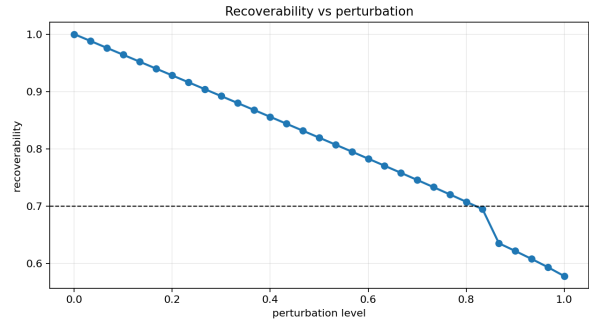
Line	Structure	k_{star}	Visible failure	Gap	Robustness
A	Gene network	0.7000	0.9333	0.2333	PASS
B	Cylindrical lattice	0.8333	0.8667	0.0333	PASS
C	3D tissue	0.8000	1.0000	0.2000	PASS
D	3D tissue through time	0.8667	0.9667	0.1000	PASS

The ladder is deliberately incremental. Line A tests an abstract regulatory system. Line B tests a cylindrical biological scaffold. Line C tests spatial multicellular stability. Line D tests developmental trajectory coherence. The success condition remains the same: recoverability collapse appears before visible failure.

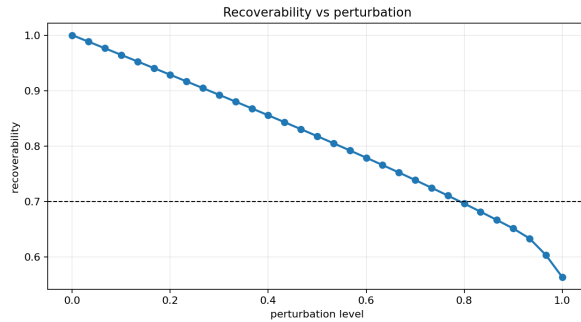
9 Recoverability Plots



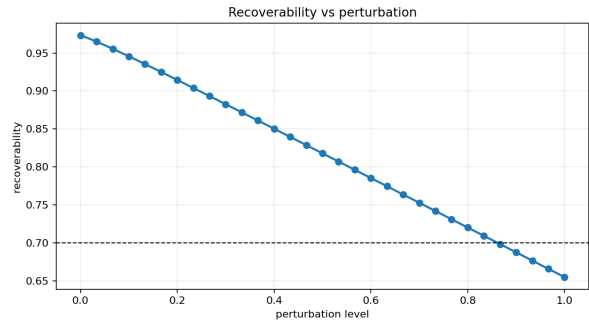
Line A: gene network



Line B: microtubule-inspired lattice



Line C: 3D tissue



Line D: 4D morphogenesis

Figure 1: Recoverability curves for Biology Lines A–D. Each curve is generated by the line-specific runner in `experiments/biology/`.

10 Reproduction Commands

All commands are run from the repository root.

```
python experiments/biology/gene_network_demo/run_gene_network_demo.py
python experiments/biology/microtubule_lattice_demo/run_microtubule_lattice_demo.py
python experiments/biology/tissue_3d_demo/run_tissue_3d_demo.py
python experiments/biology/morphogenesis_4d_demo/run_morphogenesis_4d_demo.py
```

In the local Codex environment used for this release, the demos were verified with:

```
uv run --with numpy --with matplotlib python <runner>
```

11 Conclusion

HAOS-IIP Biology Line A–D establishes a clean first biology ladder of toy recoverability diagnostics. Across four increasingly structured systems, HAOS-style proxy observables detect recoverability loss before visible failure:

1. regulatory interaction failure in a synthetic gene network;
2. lattice-support failure in a microtubule-inspired cylinder;
3. spatial-coherence failure in a 3D tissue-like structure;
4. developmental-coherence failure in a 3D structure evolving through time.

The safe public statement is:

HAOS-IIP Biology Line A–D v0.1 demonstrates early recoverability-loss detection in four toy biological interaction systems: a synthetic gene regulatory network, a microtubule-inspired lattice, a toy 3D multicellular tissue structure, and a toy 4D morphogenesis system.

Nothing stronger is asserted.